

Structured Knowledge Base for RAG: mhGAP Mental Health Intervention Protocols

1. Knowledge Base Architecture & Strategic Implementation

In resource-constrained global health settings, the transition from static PDF manuals to structured, modular "Knowledge Chunks" is a strategic imperative for clinical safety. This architecture is designed specifically for Retrieval-Augmented Generation (RAG) AI systems, ensuring that non-specialized health workers receive immediate, high-fidelity guidance without the latency or "noise" of traditional documentation. By atomizing the WHO mhGAP Intervention Guide into discrete units—defined by trigger symptoms, precise diagnostic thresholds, and scripted dialogue—we minimize the risk of AI hallucinations and ensure that clinical decision-making remains grounded in evidence-based protocols. For a health worker in a rural clinic, this structural logic provides the "So What?" layer: it transforms complex clinical guidelines into a responsive, life-saving tool that addresses the 75% treatment gap in low- and middle-income countries.

2. Core Module: General Principles of Care (GPC)

The General Principles of Care constitute the ethical and communicative bedrock of the mhGAP system. Beyond technical diagnosis, these principles ensure that care is delivered within a human rights framework that prioritizes dignity, autonomy, and the reduction of stigma.

[Topic] Communication and Relationship Building

[Trigger Symptoms] Distressed, disclosure of private/distressing information (e.g., sexual assault, self-harm), lack of understanding, cultural or gender differences, difficulty understanding health status.

[Clinical Guidance]

Ensure communication is clear, empathic, and sensitive to age and culture.
Use simple, non-technical language to explain health status.
Respond to disclosures of sensitive information with heightened privacy and sensitivity.
Ask for the person's own understanding of their condition.
Communicate realistic hope for recovery and better functioning.

[Therapist Dialogue] "I am here to listen and support you in a safe, private space. I want to understand what you feel is happening to your health. Please know that many people feel this way, and there are effective treatments that can help you recover."

[Topic] Protection of Human Rights and Dignity

[Trigger Symptoms] Stigma, discrimination (sex, religion, social origin), lack of consent, marginalization, risk of institutionalization.

[Clinical Guidance]

Promote autonomy and independent living in the community; discourage institutionalization.
Ensure care is free from discrimination based on race, sex, or indigenous origin.
Obtain free and informed consent before initiating any treatment.
Maintain strict confidentiality and respect the right to privacy.
Involve children/adolescents in decisions according to their evolving capacity.

[Therapist Dialogue] "Your privacy is essential. Before we start, I will explain exactly what this treatment involves so you can decide if you agree to it. You have the right to be involved in every decision about your care."

3. Clinical Module: Moderate-Severe Depression (DEP)

Depression is a leading cause of the global disease burden. Strategic management requires distinguishing moderate-severe depression—which demands clinical intervention—from the sadness associated with normal bereavement or adversity.

[Topic] Assessment of Moderate-Severe Depression

[Trigger Symptoms] Low energy, fatigue, sleep problems, appetite changes, persistent sad mood, irritability, loss of interest/pleasure, multiple medically unexplained aches and pains.

[Clinical Guidance]

Duration Rule: Symptoms must be persistent for at least 2 weeks.

Diagnostic Threshold: Requires at least 2 core symptoms (depressed mood, interest loss, decreased energy) AND at least 3 additional features (reduced concentration, guilt, bleak view of future, suicidal ideas, disturbed sleep/appetite).

Functioning Rule: The person must have significant difficulty in day-to-day work, school, or social activities.

Rule Out: Check for prior manic episodes (Bipolar) and recent bereavement (adjustment period).

[Therapist Dialogue] "I've noticed you've been feeling very tired and have lost interest in your usual activities. Has this been happening for most of the day, every day, for at least two weeks? Are these feelings making it hard for you to work or take care of things at home?"

[Topic] Psychosocial Management of Depression

[Trigger Symptoms] Unrealistic negative opinions, social withdrawal, inactivity, feelings of unworthiness, thoughts of self-harm.

[Clinical Guidance]

Conduct psychoeducation: explain that treatment takes weeks to work and that depression is common.

Initiate "behavioral activation": encourage resuming social activities and regular physical activity.

Address current stressors (abuse, neglect) with community resources.

Advise on regular sleep cycles (avoiding oversleeping or sleep deprivation).

Safety Note: DO NOT manage depression with ineffective treatments like vitamin injections.

[Therapist Dialogue] "It is common to have negative thoughts when you are depressed. Even if it doesn't feel enjoyable right now, trying to take a short walk or visiting a neighbor can help you start to recover. It usually takes a few weeks to feel the full benefit of treatment, so we will take this step by step."

4. Clinical Module: Psychosis (PSY)

Managing acute psychosis is a strategic priority due to the extreme vulnerability of patients to human rights violations and social exclusion.

[Topic] Identification of Acute and Chronic Psychosis

[Trigger Symptoms] Hallucinations (hearing voices), delusions (firmly held false beliefs), incoherent or irrelevant speech, disorganized behavior, extreme self-neglect, unkempt appearance, agitation.

[Clinical Guidance]

Differentiate acute from chronic (symptoms > 3 months).

Rule out delirium from medical conditions (e.g., malaria, sepsis, head injury).

Rule out substance intoxication or withdrawal.

Validate the patient's rights; people with psychosis are at high risk of neglect and abuse.

[Therapist Dialogue] "I can see that what you are experiencing feels very real to you, and it must be very distressing. You have the right to feel safe and be treated with respect. We are going to work together to find a treatment that helps these experiences become less intrusive."

[Topic] Pharmacological Management & Monitoring

[Trigger Symptoms] Acute psychotic episode, medication side effects, inner restlessness (akathisia), muscle stiffness (dystonia), non-adherence.

[Clinical Guidance]

Dosing Rule: "Start low, go slow." Use the lowest effective dose.

First-line: Offer oral Haloperidol or Chlorpromazine.

Safety Monitoring: Monitor for Extrapyrimal Side Effects (EPS).

Emergency Warning: Recognize Neuroleptic Malignant Syndrome (NMS)—characterized by rigidity, fever, and high blood pressure—as a life-threatening emergency.

Long-term Risk: Monitor for Tardive Dyskinesia (involuntary movements of face/trunk).

Adherence: Continue medication for 12 months after remission for acute psychosis.

[Therapist Dialogue] "This medicine helps your mind stay clear. We will start with a small amount. If you feel restless or your muscles feel stiff, tell me immediately. It is important to keep taking this even when you feel better to prevent the symptoms from returning."

5. Clinical Module: Bipolar Disorder (BPD)

Bipolar Disorder presents a significant economic and social cost in impoverished communities due to manic-driven reckless behavior and the high risk of relapse.

[Topic] Recognition of a Manic State

[Trigger Symptoms] Elevated, expansive, or irritable mood; decreased need for sleep; reckless behavior (excessive spending, sexual indiscretion); inflated self-esteem; extreme talkativeness; distractibility.

[Clinical Guidance]

Duration Rule: Symptoms must last for at least 1 week and interfere significantly with functioning.

Diagnostic Rule: People who experience only manic episodes (without depression) are classified as having Bipolar Disorder.

Carer Involvement: Carers are essential because patients in a manic state often lack insight and enjoy the euphoria.

Rule Out: Manic symptoms associated with drug intoxication.

[Therapist Dialogue] "You seem very energetic and haven't needed much sleep. While you might feel very good right now, I want to make sure we keep you safe so this energy doesn't lead to decisions that cause problems for you or your family later."

[Topic] Maintenance Treatment & Lifestyle Modification

[Trigger Symptoms] Irregular sleep, mood swings, history of 2+ acute episodes, mood log entries showing irritability/euphoria.

[Clinical Guidance]

Hard Stop: DO NOT prescribe Lithium if clinical and laboratory monitoring (renal, thyroid, blood levels) is unavailable.

Lifestyle: Emphasize a regular sleep cycle (same bedtime every night) and avoidance of alcohol/substances.

Medication: Maintenance with mood stabilizers (Lithium, Valproate, or Carbamazepine) should last at least 2 years after the last episode.

Monitoring: Use mood logs to track early signs of relapse.

[Therapist Dialogue] "To stay healthy for the long term, a regular sleep schedule is vital. Using this mood log every day will help us see if your mood is starting to swing so we can act early to prevent a relapse."

6. Emergency & Clinical Module: Epilepsy / Seizures (EPI)

Strategic management of seizures requires a sharp distinction between life-saving emergency intervention and chronic maintenance.

[Topic] Emergency Management of Acute Convulsions

[Trigger Symptoms] Convulsing now, unconscious, stiffness/rigidity, jerking movements, tongue bite, incontinence.

[Clinical Guidance]

Airway/Safety: Check airway; place the person on their side to prevent aspiration.

Environmental Safety: Move the person away from fire or sharp objects.

Strict Prohibition: DO NOT put anything in the mouth; DO NOT leave the person alone; DO NOT give i.m. Diazepam.

Acute Medication: Give i.v. Diazepam (10mg slowly) or rectal Diazepam. Do not exceed two doses.

[Therapist Dialogue] (To bystander) "Stay calm. Help me turn them onto their side. Do not put anything in their mouth. I am giving them medicine to stop the shaking. Stay with them until they are fully awake."

[Topic] Status Epilepticus Identification

[Trigger Symptoms] Convulsions lasting > 30 mins, or repeated seizures without recovery of consciousness in between.

[Clinical Guidance]

Priority: This is a medical emergency. **REFER URGENTLY TO HOSPITAL.**

Medication (Wait-phase): Administer Phenobarbital i.v. (10–15 mg/kg) at a rate of 100 mg/minute.

Alternative: Phenytoin i.v. (15–18 mg/kg) infused over 60 minutes.

Warning: Phenytoin is highly caustic; it requires a high-quality i.v. line to prevent local tissue damage from extravasation.

Monitoring: Watch for respiratory depression.

[Therapist Dialogue] "This seizure has lasted too long and is now a serious medical emergency. We must get them to the hospital immediately. I am administering medicine now to protect their brain while we wait for the ambulance."

7. Cross-Cutting Module: Special Populations & Concurrent Conditions

Tailoring interventions for vulnerable groups is a strategic necessity to ensure clinical safety and prevent long-term developmental or obstetric complications.

[Topic] Management during Pregnancy and Breastfeeding

[Trigger Symptoms] Pregnant, planning pregnancy, breastfeeding, slurred speech (medication effect), injection marks.

[Clinical Guidance]

Prescribing Rule: Avoid medication as far as possible; prioritize psychosocial care.

Pharmacological Safety: If medication is essential for psychosis, use low-dose oral Haloperidol or Chlorpromazine.

Epilepsy Protocol: Avoid Valproate and polytherapy. Advise hospital delivery. Give the newborn 1 mg Vitamin K i.m. to prevent hemorrhagic disease.

Hard Stop: Avoid Lithium in pregnancy/breastfeeding if laboratory monitoring is unavailable.

[Therapist Dialogue] "Because you are pregnant, we must be extra careful. We will focus on support and talk-based therapy first. If medicine is required to keep you safe, we will use the lowest dose possible to protect your baby."

[Topic] Child and Adolescent Considerations

[Trigger Symptoms] Irritability (instead of sadness), school performance problems, bullying, excessive over-activity, impulsivity.

[Clinical Guidance]

Age Threshold (< 12): DO NOT prescribe antidepressants. Prioritize psychoeducation and stressor management.

Age Threshold (12–18): Fluoxetine is the only recommended SSRI if psychosocial care fails.

Safety "Never Event": NEVER use Amitriptyline (TCAs) for adolescents.

Monitoring: Monitor for suicidal ideas weekly during the first month of Fluoxetine treatment.

[Therapist Dialogue] "I want to hear about what's happening at school. In young people, depression can often feel like being angry or irritable. We will work with your family and teachers to help things get better for you."